MZUZU UNIVERSITY

FACULTY OF INFORMATION SCIENCE AND COMMUNICATIONS

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY

END OF SEMESTER EXAMINATION

ICT 4701: SOFTWARE ENGINEERING

TIME ALLOWED: 2 HOURS **DATE : 12th SEPTEMBER, 2017**

INSTRUCTIONS

i. This paper contains **Two**pages. Please check.
ii. There are **Two questions** in this paper. Please check.
iii. Answer **ALL** questions.
iv. Marks for each question are indicated.

Question One

a. Discuss**five** components of a design method? **(10 marks)**

b. What are the **five** benefits of testing software system components individually (unit and module testing) before assembling them into a complete system? **(10 marks)**

c. Extreme programming express user requirements as stories, with each story written on a card. List and explain **two** advantages and **two** disadvantages of this approach to requirements description. **(4 marks)**

d. You have been asked to help in the development of a new computerised call centre. Customers from Mzunibankimkhonde will ring up if they wish to change the details of their account. For instance, they might want to open a new account, close an old account, they might want to change the address of the person who owns the account and so on. Briefly explain how you might use **two** different requirements elicitation techniques to identify the potential usability problems that might arise for people working with this new system. Identify the principle strenghts and weaknesses of each technique. **(10 marks)**

e. Discuss the following Software process models:
   i. Waterfall **(2 marks)**
   ii. Evolutionary **(2 marks)**
   iii. Re-use development **(2 marks)**
   iv. Spiral **(2 marks)**

b. Discuss **three** strategic options for legacy system evolution. **(6 marks)**

c. Using a diagram discuss software evolution process. **(3 marks)**

Question Two

a. What are the **eight** important attributes that all professional software should have? **(8 marks)**

b. List **two** advantages and **two** disadvantages professional software engineer should be certified in the same way as doctors or lawyers. **(4 marks)**

c. List any **two** reasons why the productivity rate of programmers working as a pair is roughly the same as two programmers working individually. **(2 marks)**

d. Discuss the following software costing models:

i. COCOMO I. **(5 marks)**
ii. COCOMO II. **(5 marks)**

END OF QUESTION PAPER